

SQL Assessment

Section A : Concept Application

1. What is the functional difference between SELECT * and specifying column names, and when is each preferred?

A. Select gives all the data from the table. Specifying column names gives the data from the mentioned name of the columns. For

Example,

Select * from student – gives all the data from the table

Select studentname from student – given only student name column

2. Which keyword renames a column in the output, and does this alias change the actual table structure in the database?

A. “as” keyword renames a column name in the output and it does not alias change the actual table structure in the database.

3. Why does wrapping a numeric value in quotes (e.g., '5000') in a WHERE clause create a data type conflict in SQL?

A. Quotes convert values into string values.

4. Contrast the results of ORDER BY Profit DESC versus ASC when the goal is to identify the top 10 most profitable orders.

A. The result of order by profit desc – shows highest amount first.

While the order by profit asc – shows lowest first.

When the goal is to identify the top 10 most profitable order

➔ Order by profit desc limit 10

5. What is the T-SQL equivalent of the LIMIT clause in MS SQL Server, and why does syntax vary across SQL engines?

A.

6. Explain the logical execution order of a query containing SELECT, WHERE, ORDER BY, and LIMIT clauses.

A. The first order of execution will be FROM as it is mandatory to give the table name. After that WHERE which is used to identify the condition or the field selection. Then SELECT as it necessary to mention which dataset need to be selected. Then we have ORDER BY weather we want column data in ascending order or descending order. In the last LIMIT which refers to how many number of data needed to be print. So the execution order is : FROM, WHERE, SELECT, ORDER BY, LIMIT

Section B: Practical Task

--1. Execute a query to retrieve the first 20 records from the orders table to verify data ingestion.

```
select * from Superstore limit 10;
```

--2. Select Order ID, Order Date, Sales, and Profit, applying a column alias to display Sales as Total_Sales.

```
select
```

```
    Order_ID, Order_Date, Sales, Profit
```

```
from Superstore;
```

--3. Filter the dataset to isolate all high-value transactions where the Sales figure exceeds 5000.

```
select * from Superstore where sales > 5000;
```

--4. Generate a report of the top 10 most profitable orders by sorting the records by Profit in descending order.

```
select * from Superstore order by Profit desc limit 10;
```